

# Knowledge Archive

NHID-Clinical · Public Reference Edition — Vision, Governance & Regulatory Alignment



v1.2

June 2026 · CC BY 4.0

**⚠** NHID-Clinical is an early-stage open proposal by Brianna Baynard. It is not an accredited standard or regulatory requirement. This is a public-safe distillation of the project's internal knowledge archive; internal-only sections (partnership discussions, decision logs, build tasking) are excluded.

## Executive Vision

NHID-Clinical was conceived from firsthand experience in payer operations. AI voice agents began appearing in B2B healthcare payer-provider calls — calls that exchange Protected Health Information (PHI) and require human escalation paths — without any consistent disclosure, identity, or audit standard.

### Impersonation Latency

The duration of time an AI agent operates and exchanges PHI without disclosing its non-human identity. NHID-Clinical median observation: 3 turns before first disclosure attempt. This is the canonical failure mode the framework exists to prevent.

| Gap                   | Description                                                                        |
|-----------------------|------------------------------------------------------------------------------------|
| Identity Gap          | No existing standard requires AI agents to identify themselves before PHI exchange |
| NPI Authorization Gap | No cross-org NPI delegation mechanism for AI agents calling on behalf of providers |
| Audit Gap             | Call-level AI decisions are not captured in healthcare-compatible audit formats    |
| Escalation Gap        | AI agents routinely fail to communicate or honor human escalation requests         |
| Deception Gap         | Some vendors use synthetic breathing/hesitation patterns implying human presence   |

## Core Framework — The Four Controls + ATR-01

### IDG-01

Identity Disclosure Gate

AI agent must disclose it is automated before any PHI request or exchange.

### PDX-01

Pre-Data Exchange Gate

No PHI may be exchanged until IDG-01 disclosure is confirmed.

## DBC-01

### Deceptive Behavior Check

No synthetic voice artifacts or text claims implying human presence.

## EIT-01

### Escalation Implementation Test

Human escalation path must be available and honored immediately on request.

## ATR-01 · Audit Trail Requirements

Every NHID event must carry event\_id, timestamp, session\_id, actor\_id, state\_before/after, and execution\_context. Validates as a plain HL7 FHIR R4 AuditEvent (base spec v4.0.1) — no named Implementation Guide conformance is claimed.

## Governance — Version Roadmap & CAS

| Version | Description                                                  | Status               |
|---------|--------------------------------------------------------------|----------------------|
| v1.0    | Original 4 controls (IDG-01, PDX-01, DBC-01, EIT-01)         | Superseded           |
| v1.3    | Current: ATR-01 added, CTS expanded to 18 tests, CAS scoring | Current              |
| v2.0    | NHID-Auth cryptographic layer (Ed25519, delegation chains)   | Reference impl. live |
| v2.1    | Planned: STIR/SHAKEN integration, attestation registry       | Future               |

## Identity & Trust Infrastructure

NHID-Auth v2 is the cryptographic authorization layer: provider-signed, NPI-bound agent passports (Ed25519), scoped delegation chains (max 3 hops), per-agent and per-delegation revocation, and call-SID nonce binding. Reference implementation: `src/agent_identity.py`.

## Technical Architecture — Live API

| Method | Endpoint                  | Auth      | Purpose                        |
|--------|---------------------------|-----------|--------------------------------|
| POST   | /v1/demo/check            | none      | Raw NHID event → result        |
| POST   | /v1/adapters/vapi/check   | none      | Native VAPI payload → result   |
| POST   | /v1/adapters/twilio/check | none      | Native Twilio payload → result |
| POST   | /v1/conformance/check     | x-api-key | Production conformance check   |
| GET    | /health                   | none      | Liveness probe                 |

## Regulatory Alignment

| Regulatory Driver   | Requirement                        | NHID-Clinical Control    |
|---------------------|------------------------------------|--------------------------|
| CMS-0057-F          | FHIR API, 72hr turnaround, 5yr log | FHIR AuditEvent + ATR-01 |
| MACPAC May 2026     | AI transparency, human review      | EIT-01 + ATR-01          |
| State AI Laws       | Inspectable, auditable decisions   | IDG-01 + DBC-01          |
| NIST AI RMF / CAISI | Cross-org agent identity           | NHID-Auth v2             |
| HIPAA Security Rule | PHI safeguards + audit controls    | PDX-01 + ATR-01          |
| TCPA                | Automated caller disclosure        | IDG-01                   |

## NIST & CMS Reference Standards

| Standard    | Reference       | Relevance                              |
|-------------|-----------------|----------------------------------------|
| STIR/SHAKEN | RFC 8224 (IETF) | Layer 1: carrier number authentication |
| FHIR R4     | HL7 FHIR v4.0.1 | Layer 4: AuditEvent resource           |
| NIST AI RMF | NIST AI 100-1   | AI risk management framework           |

|                 |                         |                                               |
|-----------------|-------------------------|-----------------------------------------------|
| NIST SP 800-207 | Zero Trust Architecture | Cross-org authorization patterns              |
| CMS-0057-F      | 88 FR 80236             | Interoperability, FHIR API, claims turnaround |
| NPI Registry    | NPPES (CMS)             | 10-digit provider identifier                  |

## FAQ & Plain Language Guide

### Is NHID-Clinical a certification or accreditation?

No. It is a voluntary open proposal (CC BY 4.0). It provides conformance scores, not formal certifications, and has no legal force.

### Does NHID-Clinical claim FHIR Implementation Guide conformance?

No. NHID-Clinical validates audit events as plain HL7 FHIR R4 AuditEvent resources against the base spec (v4.0.1) only — it does not claim conformance to any named Implementation Guide such as IHE BALP.

### We use a realistic AI voice. Does that automatically trigger DBC-01?

No. DBC-01 evaluates explicit deceptive artifact flags and impersonation phrases — it does not analyze the audio stream directly. Voice realism alone is not a violation.

### What does a payer need to start a shadow evaluation?

Access to a sample of incoming administrative call logs, about 30 minutes of staff orientation time, and optional willingness to share anonymized findings.